K-EL/01 Split-Yoke Elbow Actuator

Concept design report • Revised against the original KR-MECH-INT-001 brief

Purpose. This report documents an assignment-stage elbow actuator concept, the prescribed static gravity calculation, preliminary drive screening, a load-path concept, trade-offs, and unresolved verification tasks. It is deliberately not presented as a production-ready, safety-certified, or physically tested mechanism.

1. Requirements and basis

Input / brief requirement	Value	Status
Payload mass and COM radius	1.0 kg at 350 mm from elbow	Specified in brief
Forearm + hand mass and COM radius	1.5 kg at 180 mm	Specified in brief
Initial configuration	Forearm horizontal; gravity-only static load	Specified in brief
Output speed	45 RPM	Specified in brief
Motor / transmission / feedback	BLDC/PMSM; planetary preferred; encoder	Specified in brief
Upper arm / forearm envelope	300 mm / 350 mm CAD envelopes	Illustrative design assumptions; not brief inputs
Preliminary torque multiplier	2.0 × static gravity torque	Engineering assumption only; not a certified safety factor

2. Static sizing and speed/power check

For a horizontal forearm, each gravity torque is T = m g r. The input masses and centre-of-mass distances are taken directly from the challenge. Link masses beyond the stated forearm/hand mass, inertia, cable forces, impacts and friction are not included.

Term	Calculation	Result
Payload torque	1.0 kg × 9.81 m/s² × 0.350 m	3.4335 N·m
Forearm/hand torque	1.5 kg × 9.81 m/s² × 0.180 m	2.6487 N·m
Total static torque	3.4335 + 2.6487	6.0822 N·m
Preliminary output target	2.0 × 6.0822	12.1644 N·m
Output angular speed	45 × 2π / 60	4.7124 rad/s
Ideal output mechanical power	12.1644 × 4.7124	57.32 W
Input speed with 91:1	45 RPM × 91	4095 RPM

The 2× multiplier is a preliminary design target only. It is not evidence that acceleration, impacts, stop loads, fatigue or an industry safety factor have been satisfied. A real motion profile and application safety requirements are needed to establish peak and RMS torque.

3. Candidate drive and evidence

Part / published candidate data	Interpretation and open verification
maxon EC-i 52, 48 V, part 579165: nominal 4470 RPM, 0.423 N·m, 4.59 A; product page labels it 180 W class.	4095 RPM input target is only 375 RPM below nominal speed (~8.4% headroom). Nominal Tω arithmetic is ~198 W vs the 180 W class label; request the applicable rating point and loaded curve. Maxon says EC-i iron-winding speed/torque characteristics are nonlinear, so the published 0.321 rpm/mNm gradient must not be extrapolated as a constant slope.
maxon GP 52 C, 91:1, candidate part 223094: 30 N·m maximum continuous output torque, 45 N·m intermittent, up to 75% efficiency, ~1° average no-load backlash.	Rated torque appears to exceed 12.16 N·m but efficiency is not guaranteed at the required load/speed/duty. Product is part of a configured combination; exact orderable motor/gearhead/encoder stack remains unfrozen.
maxon ENX16EASY06 encoder candidate: 512 CPT configuration; EPOS4 Disk 60/8 CAN candidate controller.	Encoder option and controller interface/current/thermal/regen integration require configuration confirmation. Motor-side feedback does not measure output-side compliance/backlash.
2 × SKF 6201-2RSH bearings, 12 × 32 × 10 mm; KTR ROTEX GS size 19 flexible coupling with 98 ShA spider candidate.	Envelope/rating data is not a full life or application check. Coupling hub bores, keying, service factor, stiffness, speed and misalignment must be confirmed.

Efficiency sensitivity at the output target: at assumed 40% reducer efficiency, required motor torque is about 0.334 N·m, idealized motor-side power 143.3 W and a torque-constant current screen ~3.64 A. At 30%, these become ~0.446 N·m, 191.1 W and ~4.85 A; that scenario exceeds the motor's simple nominal torque/current screen and the 180 W class label. These are screening equations, not a manufacturer torque-speed, temperature or loaded-efficiency prediction.

4. Mechanical concept and load paths

K-EL/01 uses a split-yoke bearing carrier with two spaced 6201 bearing envelopes supporting an independent nominal Ø12 mm output spindle. A KTR ROTEX GS size 19 flexible coupling candidate transmits reducer torque to the spindle; the forearm attaches through a concept flange and root-mount plate. Motor, encoder, gearhead, coupling, spindle, bearings, carrier and arm links are represented as simplified CAD envelopes, not vendor-supplied solids or detailed part geometry.

- Torque path: motor → planetary gearhead → flexible coupling → independent spindle → output flange/root plate → forearm.
- Radial-load path: forearm/root plate → spindle → two spaced support bearings → split-yoke carrier → upper-arm structure. Gearhead torque rating alone is not treated as proof of cantilever radial-load capability.
- Proposed service concept (not yet proven in CAD): preassemble spindle/bearings, seat and retain the bearing set in the carrier, align the two coupling hubs, then close the removable access cap. Detailed CAD must demonstrate bearing retention, cap retention, fastener/tool reach, cable access and collision-free servicing; the current envelope model does not verify these interfaces.
- CAD-derived envelope coordinates put bearing centres at Y=11 and 61 mm (50 mm spacing); the forearm-link centre plane is Y=111 mm, hence a 50 mm overhang beyond the rear bearing centre for the simplified reaction model.

5. Structural and backlash screens

Check	Screen result	Limit
Gravity force from stated masses	(1.0 + 1.5) × 9.81 = 24.525 N	Ignores unspecified link mass and dynamics
Overhang moment	24.525 N × 50 mm = 1226.25 N·mm = 1.22625 N·m	Idealized beam; coordinates from concept CAD
Ideal support reactions	Front/outboard 49.05 N; rear/inboard 24.525 N opposite direction	Static two-support overhung-load approximation
Nominal Ø12 mm spindle bending stress	$\sigma = 32 \times 1226.25\ \text{N·mm} / (\pi \times 12^3\ \text{mm}^3) = 7.23$ MPa	Units explicitly N·mm and mm; stress concentration/dynamic loads omitted
Nominal shear at 12.1644 N·m	$\tau = 16 \times (12164.4\ \text{N·mm}) / (\pi \times 12^3\ \text{mm}^3) = 35.85$ MPa	Smooth-shaft torsion only; keyway omitted
Combined nominal von Mises	$\sqrt{(\sigma^2 + 3\tau^2)} \approx 62.52$ MPa	Static nominal screen, not fatigue approval
Candidate material basis	17-4PH supplier minimum-yield screen ~758 MPa; yield/stress quotient ~12.1	Not a design factor: heat treatment, certificate, notch, fatigue and impact remain unknown
Backlash arc	1° at 350 mm radius ≈ 6.11 mm	Gearhead average no-load backlash; not full closed-loop output error

Explicit unit conversion: use M = 1.22625 N·m × 1000 = 1226.25 N·mm before applying a formula whose diameter is in millimetres. The stress arithmetic is a nominal screen only. The actual spindle/hub/coupling interfaces and section transitions require a detailed drawing, fit and notch/fatigue review.

6. BOM, risks and assignment questions

The separate BOM.csv lists candidate, custom concept, conditional, and open parts. At minimum the BOM contains the motor and reducer candidates, encoder/controller candidates, two bearings, flexible coupling with two compatible hubs, independent spindle, interface key, split carrier/cap/fasteners, flange/root plate, simplified arm links, 48 V supply and conditional holding brake. Hub bores and shaft details are intentionally listed as open rather than guessed.

Risk / challenge	Mitigation / evidence needed
Drive may miss speed/thermal target under load.	Obtain configured drive quote, motor torque-speed curve, loaded reducer efficiency and mission profile; calculate peak/RMS current and validate temperature on a supervised bench later.
Output shaft bending/backlash/structural compliance.	Support radial load with spaced bearings; confirm spindle section/keyway/fits, bearing life, carrier stiffness, coupling torsional stiffness and output-position error.

Critical-thinking answers

Q1 — Hypothetical: torque adequate, speed misses. First check loaded motor speed-torque curve, supply voltage, controller limits and maximum reducer input speed. Test a faster motor/winding or lower ratio only after rechecking torque reserve and thermal duty.

Q2 — Overheats in repeated movement. Static gravity sizing misses acceleration, deceleration, reversals, RMS torque, copper/iron loss, gearbox friction and heat rejection. Collect current, speed, duty, winding/housing temperature and vendor thermal data over the actual cycle.

Q3 — Gearhead torque rating vs bending. A torque rating is not a radial-load or cantilever-stiffness rating. Place two separated bearings in the output carrier and route radial force into the structure; calculate shaft stress/deflection and bearing life.

Q4 — Two missing requirements. Clarify motion profile/duty/acceleration and required output accuracy/backlash/stiffness. Also obtain impact and power-loss hold expectations before design freeze; these may become safety-critical.

Q5 — Low-cost backlash vs stiffness. For an accurate elbow, tentatively favour low backlash/stiffness if performance requires it, but decide from loaded reversal error, position budget, life, efficiency, noise and budget. The ~1° no-load figure is a warning, not the assembled accuracy.

Q6 — Assumption and validation. The 2× static multiplier may not cover dynamics or impact. Derive inertia and trajectory, calculate peak/RMS torque, review stop cases and later validate a guarded, supervised duty-cycle test with current and temperature measurements.

7. Deliverables and primary references

Included beside this report: calculation workbook and CSV exports; manufacturer-reference matrix; CAD neutral STEP, parametric builder source and FreeCAD macro; assembly/one-page concept drawing; BOM; traceability, gap, risk and verification registers; simulation/physical-test plan; supplier clarification list; 10-minute slide deck; speaker notes and Q&A preparation; technical QA report; this plan and package manifest. No physical prototype, purchasing, detailed FEA, manufacturing drawings or certification are claimed, consistent with assignment scope.

Primary official product references

maxon EC-i 52 motor, part 579165: https://www.maxongroup.com/maxon/view/product/motor/ecmotor/EC-i/579165. maxon GP 52 C gearhead, part 223094: https://www.maxongroup.com/maxon/view/product/gear/planetary/gp52/223094. maxon guidance on motor data and the speed/torque characteristic of iron-winding EC-i motors: https://support.maxongroup.com/hc/en-us/articles/360013761160-Motor-data-and-simulation. KTR ROTEX GS assembly guidance: https://www.ktr.com/fileadmin/ktr/media/Manuals/45510en000000.pdf. SKF bearing product reference and the encoder/controller references are listed in Component_Research.csv. Product links confirm catalogue candidates only; the exact drive combination is not supplier-frozen.

8. Release boundary

Release decision: suitable as a documented conceptual submission subject to evaluator interpretation; not released for manufacture or operation. Open technical gates are recorded in the companion QA and Gap Closure Plan. FreeCAD was not available in this build environment, so a native .FCStd file could not be generated here; execute the supplied macro in FreeCAD to create it. Neutral STEP is the CAD exchange deliverable.